Band-1 Hadamard Order 2468 — Miyamoto (H_616 = Double(Paley-I H_308)) + Paley-I Cross-Check; Gap 2052 Still OPEN

H_308/H_616/C_618/H_2468 PASS (Miyamoto + Paley-I); 2468 in SPK via Paley I q=2467; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Lab. `/workspace/experiments/band1/hadamard_2468_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 308, 616, and 2468 only** (verified max $|HH^\top - nI| = 0$), via **two independent paths** for order 2468 (Miyamoto primary; Paley type I cross-check). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 2468 **enters SPK** via Paley I ($q = 2467$); it is **not** a Miyamoto-only bonus. Honest SPK audit: Paley-II **FAIL** ($q = 1233$ composite); Kronecker **FAIL** ($2468 = 4 \cdot 617$, 617 not Hadamard-admissible); Paley-I **PASS**. `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 2404 on box. This increment constructs the **preferred** next Miyamoto-eligible target after $q = 601$: **order 2468** ($2468 = 4 \cdot 617$, 617 prime $\equiv 1 \pmod 4$). Seed H_{308} via Paley type I ($q = 307$ prime $\equiv 3 \pmod 4$), then $H_{616} =$ Sylvester-double(H_{308}) (equivalently Kronecker(H_2, H_{308})). Symmetric Paley conference C_{618} over prime 617. $H_{2468} =$ Miyamoto(H_{616}, C_{618}). Independently, H_{2468} also verifies via **Paley type I** with $q = 2467$ (explains SPK membership). Unit tests **25/25 PASS**. SPK next gap remains **2052** (OPEN). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2468; Miyamoto; Paley type I; Paley conference; Sylvester double; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{308} constructed and verified (Paley type I, $q = 307$). 2. H_{616} constructed and verified (Sylvester-double of H_{308}; Kronecker(H_2, H_{308}) match confirmed). 3. C_{618} constructed and verified (Paley conference, $q = 617$): $CC^\top = 617I$, zero diagonal. 4. H_{2468} constructed and verified (Miyamoto). **Also** constructed and verified via Paley type I ($q = 2467$). **Not** the Hadamard conjecture. 5. SPK: $2468 \in$ SPK (via Paley I). Paley-II and Kronecker paths for 2468 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$2468 = 4 \cdot 617$
$q = 617$	prime, $617 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{616}
Needed conference	order $618 = q + 1$
H_{308} path	Paley type I, $q = 307$ (prime $\equiv 3 \pmod 4$)
H_{616} path (used)	Sylvester-double(H_{308}) $\equiv$ Kronecker(H_2, H_{308})
Conference path	Symmetric Paley over prime 617
SPK path for 2468	Paley type I, $q = 2467$ prime $\equiv 3 \pmod 4$; order $q + 1 = 2468$
Paley-II for 2468	**REJECTED**: $q' = 1233$ not prime
Kronecker for 2468	**REJECTED**: only 4×617; $617 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{308}	Paley type I, $q = 307$	0	0.0533	**PASS**
H_{616}	Sylvester-double(H_{308})	0	0.1132	**PASS**
C_{618}	Paley conference, $q = 617$	0	0.2902	**PASS**
H_{2468}	Miyamoto(H_{616}, C_{618})	0	6.3313	**PASS**
H_{2468} P-I	Paley type I, $q = 2467$	0	9.2956	**PASS**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh + Paley-I cross-check) ≈ 16.8582 s. Compressed ± 1 matrices saved as `H308_pm1.npz`, `H616_pm1.npz`, `C618_pm1.npz`, `H2468_pm1.npz`, `H2468_paleyI_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships $\{24, 284, 508, 528, 540, 556, 568, 576, 600, 2004, 2036, 2116, 2164, 2228$ $\cup$ verified H_{308}, $H_{616} \cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 2468 enters SPK **because** Paley I applies ($q = 2467$).

Quantity	Value
Known count	880
2468 in SPK	**True** (Paley I $q = 2467$)
616 in SPK	True (Double / Kronecker of Paley I 308)
308 in SPK	True (Paley I $q = 307$)
2404 in SPK	True (prior; Paley II $q = 1201$)
2276 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284...

5. Gap 2052 (still OPEN; non-finding ONLY)

This ship does **not** fill gap 2052. Prior triage stands: Paley-I/II / Kronecker / Miyamoto-Paley-conf all **FAIL** for order 2052. Claim level for 2052 remains method triage / non-finding ONLY. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (25/25)

Test	Result
`H20_paley_I_control`	PASS
`H4_sylvester`	PASS
`paley_II_q5_H12` / `paley_II_q13_H28`	PASS
`miyamoto_H20_from_H4_conf6`	PASS
`paley_I_q11_H12`	PASS
`q617` / `n2468` / `q307` / `H308` / `H616` / `C618` formulae	PASS
`q2467` / Paley-I order-2468 formula	PASS
`paley_II_H2468_path_rejected`	PASS
`kronecker_H2468_path_rejected`	PASS
`reject_all_ones_H8`	PASS
`conference_6_paley_control`	PASS
`miyamoto_eligible_2468_arith`	PASS
`conference_618_shape_expected`	PASS
`double_paley_I_H12_to_H24`	PASS
`gap_2052_not_filled_by_this_ship`	PASS

7. Next optional targets

Miyamoto-eligible primes $q > 617$, $q \equiv 1 \pmod 4$ (sample):

q	$n = 4q$	Need H_{q-1}	On-box path notes
641	2564	640	Kronecker(4x160)
653	2612	652	no quick SPK seed noted
661	2644	660	PaleyI(q=659)
673	2692	672	Kronecker(4x168)
677	2708	676	PaleyII(q=337)
701	2804	700	PaleyII(q=349)

Preferred next. Method-attack remaining SPK gap **2052** (GS/SDS for order ~513 if literature seeds appear), **or** construct H_{2644} (Paley-I seed $q = 659$) / H_{2564} (Kronecker 4×160), **or** Track B / Object 1 await. GATE-04 remains LOCKED.

8. Non-claims (explicit)

- Full Hadamard conjecture: **OPEN**
- Clay-closed / finished-ToE: **false / not claimed**
- Gap 2052 filled: **NOT claimed** (OPEN)
- GATE-04: **LOCKED**

- RH / P≠NP / YM / Frankl / NS-solved / AC-solved: **not claimed**
- Miyamoto-only bonus for 2468: **false** (in SPK via Paley I)

9. Reproducibility

```
cd /workspace/experiments/band1/hadamard_2468_miyamoto
python3 verify_hadamard_2468.py
# expects: Unit tests 25/25; H308/H616/C618/H2468 Miyamoto PASS; Paley-I cross-check PASS
# artifacts: results.json, literature_ledger.json, SUMMARY.txt, *.npz
```

Lab path. `/workspace/experiments/band1/hadamard_2468_miyamoto/`
PDF shelf. `/workspace/DOWNLOAD_PDFs/2026-09-15_Band1_Hadamard_2468_Miyamoto.pdf`